Key Findings

Competitor Research Key Findings

Extrinsic Gameplay Breaks Flow:

Pausing combat or exploration for pop-up quiz cards (e.g., Prodigy) makes math feel like a penalty tollgate rather than a core game mechanic ("chocolate-covered broccoli").

Punitive Scoring Triggers Anxiety:

Slashing student progress scores immediately upon an incorrect answer (e.g., IXL) increases math anxiety and frustration.

Static Question Banks Cause Churn:

Pre-scripted question pools (e.g., Monster Math) lead to answer memorization and low replayability once content runs dry.

High School Market Gap:

Current gamified math platforms focus almost exclusively on early primary learners (Grades 1–6), leaving secondary students (Grades 7–11) with no age-appropriate RPG alternatives.

Key Findings

Secondary Research Key Findings (Literature & DBE Data)

Anxiety Degrades Working Memory:

Math anxiety causes a psychological stress response that actively reduces working memory capacity during problem-solving (Ashcraft, 2002; Maloney & Beilock, 2012).

Global Prevalence is High:

Over 39% of 15-year-old students globally experience severe tension or helplessness when solving math problems (OECD, 2023).

Foundational Gaps Drive Pure Math Drop-Off:

Only 34.1% of South African matric students write pure Mathematics, largely due to unaddressed arithmetic and algebra gaps accumulating from earlier primary/ intermediate grades.

Mid-Task Diagnostic Support is Essential:

Static text solutions provided after a problem is marked wrong fail to guide students through logical steps, whereas mid-task Socratic hints preserve confidence and maintain flow state.

Key Findings

Survey Research Key Findings

High Math Anxiety & Execution Blockers:

71.5% of students report severe math anxiety, with 57.1% identifying "Applying Theory to Action" knowing formulas but freezing when starting a problem on paper, as their single biggest obstacle.

Eliminate/limiting Time Pressure:

85.7% of students freeze up and panic during countdown timers, explicitly rejecting mandatory timed mechanics in favor of low-stress environments.

Holistic RPG Integration:

71.4% want math fully integrated across gameplay (casting spells/traps), rewards, and narrative progression, while 100% want the system to cover both topic reinforcement and Learning integation .

Constructive Turn-Based Feedback:

71.4% prefer turn-based battle mechanics where a failed math attack explains the mistake before the enemy counters, turning gameplay errors into learning moments.

Key Findings

Gamification Research Key Findings (Literature)

Gamification Improves Learning:

Across many studies, gamified learning clearly improved what learners learned (g = 0.49), not only how much they enjoyed it (Sailer & Homner, 2020).

Motivation & Engagement Rise:

Gamification raised learners' motivation (g = 0.36) and how actively they took part in tasks (g = 0.25), with story worlds helping most (Sailer & Homner, 2020).

Games Beat Non-Game Lessons:

Digital learning games improved learning compared with non-game teaching (g = 0.33), and design choices changed how well they worked (Clark et al., 2016).

Maths Must Be the Gameplay:

Children learned more and played about 7× longer when the maths was built into the core gameplay instead of used as a reward between questions (Habgood & Ainsworth, 2011).

Research Findings

FINDINGS

What the research showed (survey, DBE data, literature, competitors)

- 71.5% report severe maths anxiety (from my survey).
- 57.1% get stuck at “applying theory to action”: they know the formula but can’t start.
- 85.7% freeze or panic under countdown timers.
- Only 34.1% of matric candidates write pure Mathematics; the pass rate fell from 69% to 64% (DBE, 2026).
- 39% of 15-year-olds feel helpless with maths problems (OECD, 2023).
- Gamification improves learning, g = 0.49 (Sailer & Homner, 2020).
- Maths as the gameplay: ~7× longer play (Habgood & Ainsworth, 2011).

PATTERNS

What kept coming back across sources

- Anxiety → less working memory → mistakes → avoidance: the literature and the survey describe the same cycle.
- Learners struggle with execution, not with understanding the theory.
- Competitors treat maths as an interruption (Prodigy), punish mistakes (IXL) or repeat fixed questions (Monster Math).
- Feedback usually comes after the answer is marked wrong, not during the steps.
- Games teach best when the learning is built into the play, not added on top as points or rewards.

Research Findings

TENSIONS

Needs that pull against each other

- Fun vs. learning: 42.9% value fun most, 42.9% real maths learning (Q16 of survey).
- Challenge vs. anxiety: a game needs risk, but timers and penalties cause panic.
- Older learners want depth, but existing games feel childish.

OPPORTUNITIES

Gaps no current product fills

- An RPG made for Grades 7–11 with real algebra and geometry.
- Maths as the core mechanic: 71.4% want it built into spells, traps and rewards.
- Turn-based play with no timers.
- Fresh, CAPS-aligned questions so answers can't be memorised.

INSIGHTS

What this means for the design

- Guide the next step, don't just mark the answer: mid-task hints protect confidence.
- A wrong answer should teach: 71.4% want the mistake explained before the enemy acts.
- Remove time pressure so working memory is free for the maths.
- The learning has to be the game, not the toll-gate in front of it: a game you learn in, not a gamified app.